

RAAKESH MENON A

SENIOR DATA ENGINEER

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PROFESSIONAL SUMMARY

Senior Data Engineer with 4+ years of experience building enterprise data platforms, modernizing legacy ETL, and delivering cloud-based data pipelines across banking, semiconductor manufacturing, aerospace, and animal-health domains. Strong in Python, Advanced SQL, Snowflake, PySpark, Apache Spark, data migration, data modeling, data quality, validation, and distributed processing. Experienced in Azure data engineering, including Data Factory, ADLS Gen2, Databricks, Azure SQL, and modern data-platform architecture, with additional experience applying Microsoft Fabric, OneLake, and LLM-assisted automation to data engineering workflows.

CORE TECHNICAL SKILLS

Programming: Python, Advanced SQL, PL/SQL, Perl, Shell Scripting, REST API Integration, JSON, XML

Data Platforms: Snowflake, Azure Databricks, Apache Spark, PySpark, Delta Lake, Microsoft Fabric, OneLake

Data Engineering: ETL/ELT, Data Pipeline Architecture, Data Integration, Data Transformation, Data Migration, Data Warehousing, Data Modeling, Batch Processing, Incremental Processing, Workflow Orchestration, Source-to-Target Mapping, Reusable ETL Components, Data Quality, Query Optimization, Performance Optimization

Data Quality and Optimization: Data Validation, Data Profiling, Data Reconciliation, Source-to-Target Validation, Anomaly Detection, Data Quality Controls, Validation Automation, Reusable Validation Frameworks

Azure and Cloud: Azure Data Factory, Azure Data Lake Storage Gen2, Azure SQL, Azure Functions, Azure Synapse Analytics, Azure Databricks, Delta Lake, Microsoft Fabric, OneLake, Azure DevOps, Azure Boards, Azure Repos, Azure Pipelines, Cloud Data Engineering, Cloud Data Integration, Azure Blob Storage, SharePoint, Microsoft Graph, iPaaS, Azure Logic Apps

AI & Intelligent Data Engineering: Anomaly Detection, MLflow, LLM Integration, LLM-Assisted Data Engineering, Fabric Agents

BI and Engineering Delivery: Power BI, DAX, Power Query, Tableau, Looker Studio, Advanced Excel, KPI Dashboard Development, Enterprise Analytics, Reporting Automation, Git, CI/CD, Agile Methodology, SDLC, Release Coordination, Deployment Coordination

PROFESSIONAL EXPERIENCE

Senior Systems Engineer / Senior Data Engineer

Apr 2023 – Present

Tata Consultancy Services (TCS) — Bengaluru, India

- Engineered data engineering for ~2 TB of semiconductor manufacturing data across 300+ BOM and product-hierarchy schemas, designing ingestion and transformation pipelines integrating SAP and Teamcenter data using Azure Data Factory, ADLS Gen2, Databricks, Snowflake, Python, SQL, and PySpark.
- Automated validation, reconciliation, profiling, anomaly detection, KPI scoring, and reporting workflows, reducing validation and reconciliation effort by 95% and supporting KPI dashboards for approximately 250 users through Python, SQL, Spark processing, Snowflake, and Power BI.
- Designed enterprise financial data migration pipelines processing ~500K records across 5 source systems, implementing transformation, validation, reconciliation, and curated Snowflake loading using Python, SQL, PySpark, and Spark.
- Built 20+ Spark and ETL workflows incorporating migration controls, reconciliation, orchestration, query optimization, and data-quality validation for curated enterprise datasets.
- Engineered an Azure-based migration architecture spanning 10 source systems and 300+ schemas, integrating DB2, Snowflake, files/scripts, and IBM Watson Studio through Azure Data Factory, ADLS Gen2, Azure SQL, Azure Functions, Databricks/Spark, and Delta Lake.
- Optimized migration pipelines through partitioned Spark processing, SQL optimization, reusable transformation patterns, and data-quality controls, delivering curated datasets into Snowflake/Fabric for Power BI analytics supporting 1,000+ reporting consumers.
- Designed reusable ingestion and transformation patterns with source-to-target mappings, incremental and batch processing, validation/reconciliation controls, partition-aware Spark execution, optimized SQL transformations, and curated Snowflake loading for downstream analytics.

- Delivered production-oriented pipelines using Python, Advanced SQL, Snowflake, PySpark, Apache Spark, reusable ETL components, query optimization, Git, Azure DevOps, CI/CD, Agile/SDLC practices, workflow orchestration, and data-quality controls.
- Built an LLM-augmented ETL modernization framework that translated legacy Perl/SQL workloads into Python using AST-based parsing, Claude API, Airflow, PySpark, Snowflake, Great Expectations, and automated behavioral-equivalence testing; achieved 85% automated translation and reduced manual conversion effort from 40 hours to 4 hours per script, generating 129 regression cases across the migration corpus.

Systems Engineer / Data Engineer

Dec 2021 – Mar 2023

Tata Consultancy Services (TCS) — Bengaluru, India

- Modernized legacy financial-services ETL by converting 800+ Perl scripts to Python and migrating batch-driven processing to Talend, integrating IBM DB2 extraction, reusable transformation logic, validation, and Power BI delivery.
- Reduced ETL processing time by 90% through dependency reduction, reusable Python components, query optimization, workflow automation, and batch migration, while developing 50+ reusable components for repeatable data engineering delivery.
- Engineered Python and SQL ETL workflows for aerospace manufacturing data, processing approximately 100K records across 5+ source systems including IBM DB2, modernizing Legacy ETL Perl based processing for downstream Power BI and Excel reporting.
- Supported 70–80+ ETL workflows and 100+ reports, improving reporting performance through reusable Python components, SQL optimization, standardized transformations, validation controls, and workflow automation.
- Developed Python and SQL ETL pipelines for animal-health datasets, integrating SAP HANA data into reusable processing workflows and supporting approximately 10 million records across 20+ ETL workflows using Snowflake and enterprise reporting patterns.
- Reduced a measured manual process from 48 hours to approximately 2 hours, an improvement of up to 95%, and improved processing performance by 80% through Python automation, SQL validation, reusable transformation components, source-to-target reconciliation, and scalable Snowflake processing.
- Designed reusable data-processing workflows across structured and semi-structured datasets using Python, Advanced SQL, Snowflake, PySpark, Apache Spark, Talend, data modeling, source-to-target validation, profiling, reconciliation, batch processing, and performance optimization.
- Supported end-to-end data engineering delivery through Git, CI/CD, Agile/SDLC practices, workflow orchestration, query tuning, incremental processing, reusable ETL frameworks, monitoring, release coordination, and Power BI-ready curated datasets.

ACHIEVEMENTS

- Earned multiple stakeholder appreciation awards and internal leadership for delivering 10+ high impact data engineering and analytics solutions improving operational efficiency and reporting reliability.
- Conducted 10+ technical knowledge sharing and upskilling sessions on Python, SQL, and ETL best practices for 25+ engineers, strengthening team capability in analytics engineering.

EDUCATION

Bachelor of Engineering | Anna University, Chennai, India

2016 – 2020

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate
- IBM Data Science Professional Certificate
- Google Business Intelligence Professional Certificate
- Generative AI with Large Language Models | DeepLearning.AI